sprinkling of stars with a pattern of concentric shells, a pattern produced by the gravitational lensing.

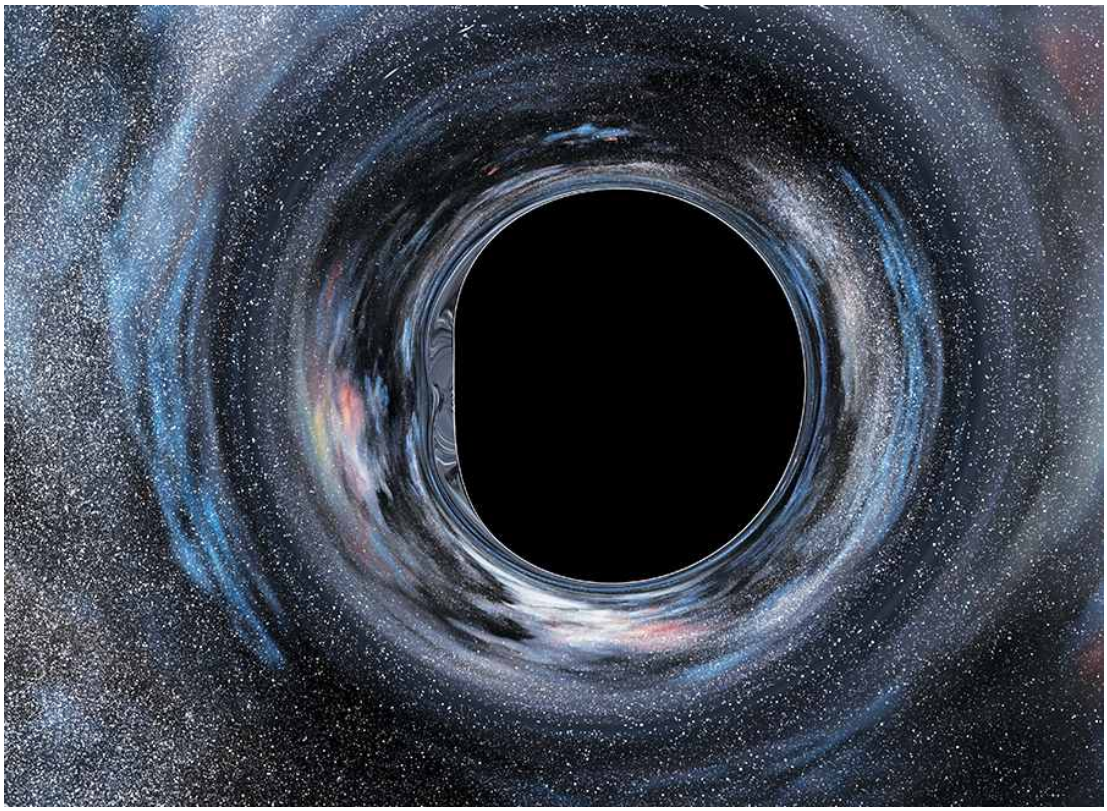


*Fig. 8.1.* The gravitationally lensed pattern of stars around a rapidly spinning black hole such as Gargantua. When seen from far away, the shadow's angular diameter, measured in radians, is 9 Gargantua radii divided by the observer's distance from Gargantua. *[From a simulation by the Double Negative visual-effects team.]*

As the camera orbits around Gargantua, the field of stars appears to move. This motion combined with the lensing produces dramatically changing patterns of light. The stars stream at high speed in some regions, they float gently in others, and they're frozen in still other regions; see the film clip on this book's page at Interstellar.withgoogle.com.

In this chapter I explain all these features, beginning with the shadow and its ring of fire. Then I describe how the black-hole images in *Interstellar* were actually produced.